Computers and Geotechnics 199 (2026) 108362

Contents lists available at ScienceDirect

Computers and Geotechnics

journal homepage: www.elsevier.com/locate/compgeo

Research paper

GRAI3: Generalizable soil particle size distribution from smartphone images☆

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ARTICLE INFO

Keywords:
Soil particle size distribution
Deep learning
Smartphone imaging
Computer vision
Geotechnical characterization
Transfer learning

ABSTRACT

We present GRAI3, a device-agnostic web application for generalizable prediction of soil particle size distribution from standardized smartphone images. GRAI3 couples a streamlined acquisition protocol using a photo-box with a deep learning pipeline based on EfficientNetV2-S, trained on an expanded, diverse dataset. The method models the cumulative particle size distribution with a two-parameter Weibull function, inferring its scale and shape directly from an image. To achieve device-agnostic performance, GRAI3 automatically reads camera EXIF metadata, retrieves per-device pixel density from a calibration table or estimates pixel density when the device is unknown based on the typical angle of view. Images are normalized to a target pixel density and centre-cropped to a fixed resolution before inference, reducing covariate shift across cameras. A lightweight verification model checks whether the uploaded image depicts soil and adheres to photo-box capture requirements. The web application performs real-time inference and provides a one-click PDF report, embedding the original photo, PSD curve and computed parameters. Monte Carlo dropout was implemented to provide a measure of confidence in the model's predictions. Comparison with a SAM-refined, U-Net-based segmentation pipeline shows that GRAI3 remains robust with fine textures. Validation on an expanded dataset and cross-device scenarios indicates improved generalizability relative to prior versions, attributable to the upgraded backbone and group cross-validation. GRAI3 thus offers a practical, reproducible workflow for soil granulometry estimation using commodity smartphones, supporting rapid screening, documentation and reporting in geotechnical and environmental applications.

1. Introduction

Soil particle size distribution (PSD) underpins many geotechnical decisions because it governs pore architecture and packing, which in turn control strength and flow. Well-graded soils typically achieve higher density and interlocking, leading to larger friction angle than uniformly graded counterparts. Hydraulic conductivity is closely linked to PSD percentiles (e.g., Hazen's approximation $k \propto D_{10}^2$ for clean sands Hazen, 1911) and fines content, which can constrict pore voids and reduce permeability. Filter design relies on PSD-based criteria to prevent piping while ensuring drainage, for example $D_{15,\text{filter}} \leq 4D_{85,\text{base}}$ for retention and $D_{15,\text{filter}} \geq 4D_{15,\text{base}}$ for permeability (Head, 2006). PSD also shapes compaction behaviour: well-graded soils tend to attain higher maximum dry unit weights as smaller particles fill voids between larger ones, while excess fines raise optimal water content and can hinder granular rearrangement (Daddow and Warrington, 1983; Goldsmith et al., 2001; Lambe, 1958).

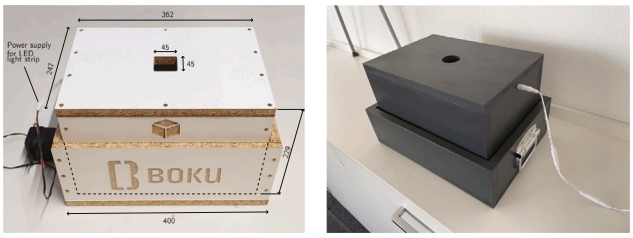
Conventional sieve and hydrometer tests are cumbersome and slow. Turnaround times are often hours to days once you factor in drying, splitting, washing, sieving and hydrometer runs with temperature corrections. They require dedicated equipment (nested sieve stacks, shaker, hydrometers, cylinders), consumables (dispersants, wash water) and trained technicians to manage sample preparation (deflocculation, dispersion, removal of organics) and interpret corrections. Alternative approaches, such as the Camsizer (Microtac, 2025), use dynamic image analysis, laser diffraction (Mastersizer Malvern Panaytical, 2025), sensor-based free-fall sorting (colour, NIR, laser and electromagnetics TOMRA, 2025) or image segmentation (Wipfrag WipWare, 2026).

Recently, computer vision has emerged as a practical route to obtain PSD from soil images. Approaches span photos of borehole samples (Granitzer et al., 2025), lab samplers and bench-top setups (Boulos et al., 2025) and soils arranged in trays (St-Cyr et al., 2025b). Some scholars have also attempted to classify the granulometry of river

☆ Funded by the European Union under the MSCA Staff Exchanges project 101182689 Geotechnical Resilience through Intelligent Design GRID. Views and opinions expressed are however those of the author(s) only and do not necessarily reflect those of the European Union or European Research Executive Agency (REA). Neither the European Union nor the granting authority can be held responsible for them.

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https://doi.org/10.1016/j.compgeo.2026.108362
Received 4 March 2026; Received in revised form 4 May 2026; Accepted 10 June 2026
Available online 17 June 2026
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(a) BOKU's photo-box [24, 26]. (b) HPC's photo-box.

Fig. 1. Photo-boxes used for standardized image acquisition.

Table 1
Summary of data sources and contributors to the training dataset.

Series	Contributor	Soil provenance and type
GRAI1	BOKU University	Mainly Vienna area, Nepal
GRAI2	HPC	German soils
GRAI3	BOKU	Poorly graded soils
GRAI3	HPC	German soils
GRAI3	Tauchmann	West Austrian soils

beds from photographic images, extending these methods to fluvial environments (Buscombe, 2020; Lang et al., 2021). Some workflows also incorporate corrections for confounding factors such as water content (Boulos et al., 2025), improving robustness across real-world conditions. Importantly, several datasets were released to encourage benchmarking (Soranzo, 2025a; St-Cyr et al., 2025a).

In previous studies, we developed convolutional neural networks for image-based PSD estimation. The first study, GRAI1 (Soranzo et al., 2025), demonstrated that the full PSD curve can be predicted from close-range soil images by regressing a two-parameter Weibull function via a MobileNet (Howard et al., 2017) backbone. GRAI2, a subsequent version of the CNN (Soranzo, 2025c), targeted broader deployment and robustness by incorporating new data into training, applying additional data augmentation (hue and saturation adjustments) to better span illumination and colour variability and adopting 5-fold cross-validation to stabilize estimates across subsets.

Generalizability emerged as the central challenge in the previous iterations. A sensitivity analysis (Huepe-Marinovic, 2025) revealed that the model underperformed on very steep PSDs. Compounding this, minimally altered replicates of the same soil leaked into the test set, obscuring true out-of-sample error. Augmentation was performed across only the training and not the test split, but it was also applied prior to the validation split, creating distributional overlap. The web applications (Soranzo, 2024, 2025b) supported only pre-calibrated smartphones, offering no path for unknown devices. Finally, pictures of non-soils were accepted without warning.

To address these challenges, we expanded the training set to include poorly graded soils. We eliminated leakage by adopting group cross-validation. We adapted the augmentation protocol by applying transformations only after the train/validation split. Recognizing persistent underperformance with MobileNet, we upgraded the backbone to EfficientNetV2-S (Tan and Le, 2021). We expanded the roster of supported smartphones and introduced a device-agnostic path for unknown cameras: the web application reads EXIF metadata and, if the camera is unknown, estimates pixel density from optical parameters (e.g., focal length and angle of view). To prevent non-soil images from passing silently, we trained a lightweight classification CNN to gate uploads before PSD inference.

GRAI3 substantially improves generalizability relative to prior versions. Across unseen groups, we achieved $R^2 = 0.963$ and $R^2 = 0.880$ for the predicted Weibull parameters, with mean absolute percentage errors of 23.2% and 18.6%, respectively. These results reflect robust performance under harsher, leakage-resistant evaluation.

Table 2
Smartphones and tablets used to train the model and additional devices that are automatically recognized for inference.

Devices used for training	Resolution (px)	PPM	AOV (°)
iPad Pro	4032 × 3024	15.0	65.4
iPhone 14	4032 × 3024	13.9	69.1
iPhone 16	5712 × 4284	19.5	69.7
Motorola Edge	4000 × 1800	11.5	79.3
Samsung Galaxy A32	4000 × 2250	12.9	72.8
Samsung Galaxy A32	8000 × 6000	25.8	72.8
Samsung A52	9248 × 6936	26.3	79.8
Additional devices for inference	**Resolution (px)**	**PPM**	**AOV (°)**
Huawei p9 Lite 2017	3968 × 2976	13.7	69.0
iPhone 14 Pro	4032 × 3024	12.3	76.0
iPhone 15 Pro	4032 × 3024	12.8	73.8
iPhone 16 Pro	3024 × 4032	12.6	74.5
Motorola Edge 60 fusion	2304 × 4096	12.5	76.1
Motorola G7+	4608 × 3456	18.3	61.9
Motorola G30	4624 × 3472	17.6	64.1
Samsung A52	9248 × 6936	26.3	79.8
Samsung Galaxy S23	4000 × 3000	12.5	74.7
Samsung Galaxy S24	4000 × 3000	12.2	75.8
Vivo X60	4000 × 3000	15.4	63.3
Xiaomi 15	3072 × 4096	12.5	75.8
Xiaomi A2	4000 × 3000	15.7	62.5

2. Materials and methods

2.1. Data acquisition

Images were captured with smartphone cameras inside a photo-box, originally with the model shown in Fig. 1(a). Soil samples were homogenized in a cement mixer, oven-dried and then dry-pluviated into the photo-box tray. Because the box dimensions are publicly available (Fig. 2), collaborators independently built compatible boxes (Fig. 1(b)) and contributed data from various geotechnical projects. The expanded dataset for GRAI3 incorporates samples from BOKU University, as well as contributions from industrial partners (Table 1). These samples cover a broad geographic and compositional range, including soils from Austria, Germany and Nepal. This multi-source acquisition, using replicated but independently constructed setups, enhances the diversity of imaging conditions and improves the model's generalizability. Photos were taken through a 45×45 mm aperture at a fixed stand-off of 210 mm from the soil surface. The devices used are listed in Table 2.

Because cameras differ in angle of view (AOV), different smartphones have different pixel densities (pixels per millimetre, PPM). The pixel density is defined with the following expression:

$$\mathrm{PPM} = \frac{w}{b} \tag{1}$$

where w is the width of the image in pixels and b is the width of the photographed area in mm. The AOV is defined according to Fig. 3 and can thus be obtained with the expression

$$\tan\left(\frac{\mathrm{AOV}}{2}\right) = \frac{b}{2h} \tag{2}$$

Hence, the pixel density is also equal to

$$\mathrm{PPM} = \frac{w}{2h\tan\left(\frac{\mathrm{AOV}}{2}\right)} \tag{3}$$

For unsupported phones, a first-order calibration can be obtained by assuming AOV $\approx 70°$ given the fixed $h = 210$ mm imaging distance. This 70° has been obtained by rounding the average angle calculated from Table 2. Hence, we can obtain b from Eq. (2) and the PPM from Eq. (1) becomes:

$$\mathrm{PPM} = \frac{w}{2 \cdot 210 \cdot \tan 35°} = \frac{w}{294.087} \tag{4}$$

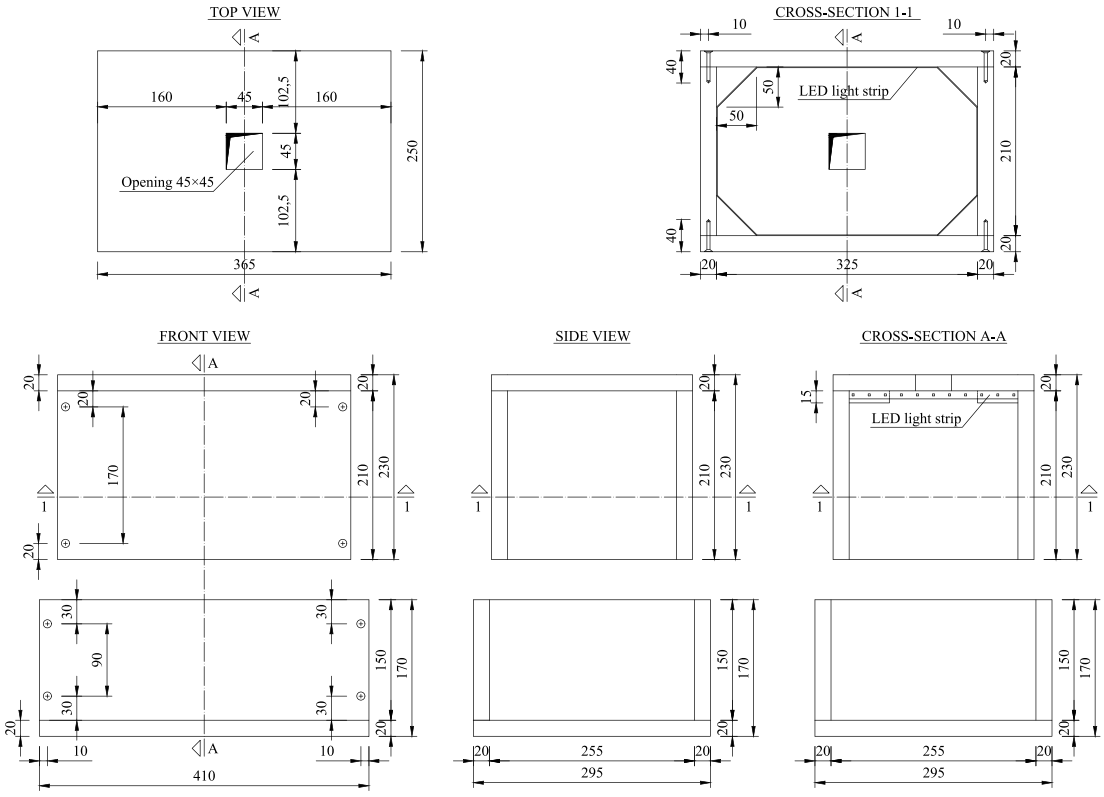


Fig. 2. Dimensions of the photo-box (Soranzo, 2026).

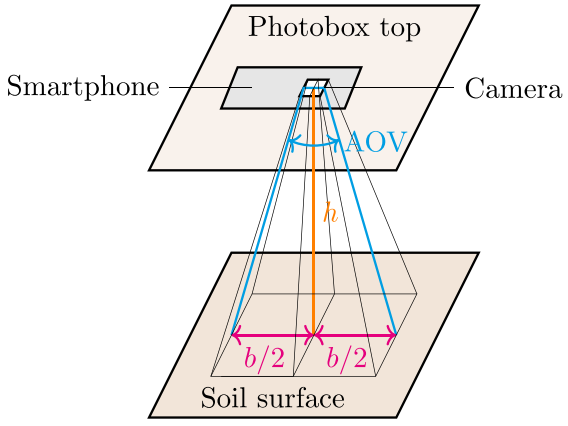


Fig. 3. Definition of the angle of view.

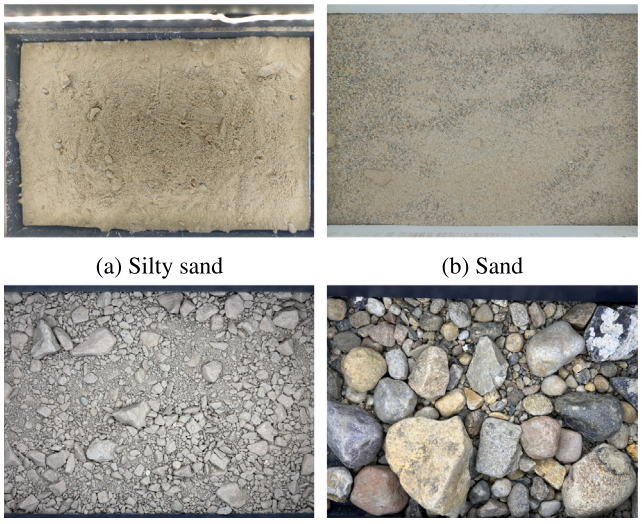


Fig. 4. Representative original smartphone images of soil samples from different data sources.

The pixel density of supported phones (Table 2) was previously calibrated; based on this density, images are resized to a target of 4.6 PPM. Images are centre-cropped into 400 × 400 px squares.

To reduce bias and spatial correlation, multiple images of each soil were acquired while reshaping the surface between shots using simple tools. In total, 321 images were collected from 58 distinct soil samples. Representative images of the soil samples are displayed in Fig. 4. The soil samples exhibit average clay, silt, sand and gravel contents of 3.1%, 11.7%, 47.7% and 36.4%, respectively. The measured particle size distribution curves for all samples are shown in Fig. 5, highlighting the contributions from different generations of the dataset (GRAI1, GRAI2 and the new data in GRAI3).

PSD curves were obtained by sieving for coarse fractions and by the hydrometer sedimentation method for fine fractions. The coefficients of uniformity and curvature varied between $C_U \in [1.7, 275.0]$ and $C_C \in [0.5, 5.9]$, representing a wide range of soil gradations from poorly to well-graded materials.

The PSDs were fitted with the two-parameter Weibull distribution:

$$F(d \le d_i) = 1 - e^{-(d_i/b)^c} \tag{5}$$

where d is the particle diameter, d_i is a specific diameter threshold and b and c are fitting parameters representing the scale and shape of the

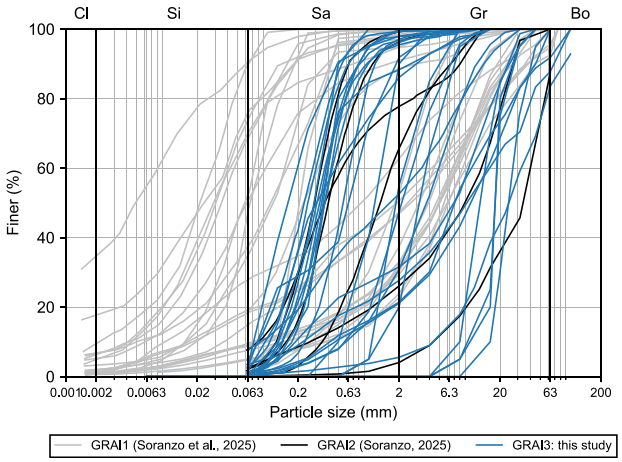


Fig. 5. Measured PSD curves for the 58 soil samples used in the study, encompassing a wide range of gradations and highlighting the original and newly added data.

Table 3
Evaluation of fallback PPM estimation for unseen devices.

Evaluated devices	PPM_{emp}	AOV_{emp}	$PPM_{fallback}$	APE
iPhone 13	5.796	67.380	5.804	0.1%
iPhone 17	19.543	67.380	19.423	0.6%
Redmi Note 11 Pro 5G	12.919	65.470	13.601	5.3%

distribution, respectively. As explained in Soranzo et al. (2025), b and c are related to a characteristic particle size and to the coefficient of uniformity:

$$d_{50} = 0.69^{1/c} \cdot b \tag{6}$$

$$C_U = 8.70^{1/c} \tag{7}$$

Since the parameters were comprised in the ranges $b \in [0.030, 39.561]$ and $c \in [0.392, 3.284]$, the first term in Eq. (6) varies between 0.388 and 0.893 and thus the mean diameter is mainly determined by b rather than by the curvature. For the same reason, since b shows greater variability than c, the natural logarithm of b is used as the output variable to stabilize the variance during model training. In doing so, the range of $\ln(b)$ becomes $[-3.508, 3.678]$.

2.1.1. Evaluation of fallback PPM estimation for unseen devices

When a camera model is not present in the calibration table, our system uses a device-agnostic fallback based on a typical AOV to estimate PPM. To assess this fallback under realistic user conditions, we held out three smartphone models from the calibration dataset and collected test measurements using the same imaging box used in deployment.

For each held-out device, we measured the pixel length w of a known physical length b. We then measured the line length in pixels; we converted this measurement to an empirical PPM value as

$$PPM_{emp} = \frac{w}{b} \tag{8}$$

We compare PPM_{emp} to the fallback PPM estimate ($PPM_{fallback}$). Table 3 reports per-device absolute percentage error

$$APE = \frac{|PPM_{emp} - PPM_{fallback}|}{PPM_{emp}} \tag{9}$$

Across the three held-out smartphones, the maximum fallback APE is about 5%, demonstrating that the typical AOV fallback yields robust normalization in practice.

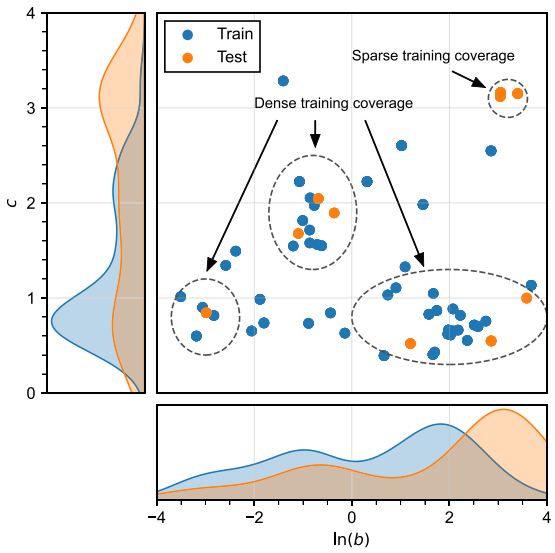


Fig. 6. Pair-plot and kernel density estimation (KDE) of the Weibull parameters illustrating the train–test split and regions of sparse training coverage.

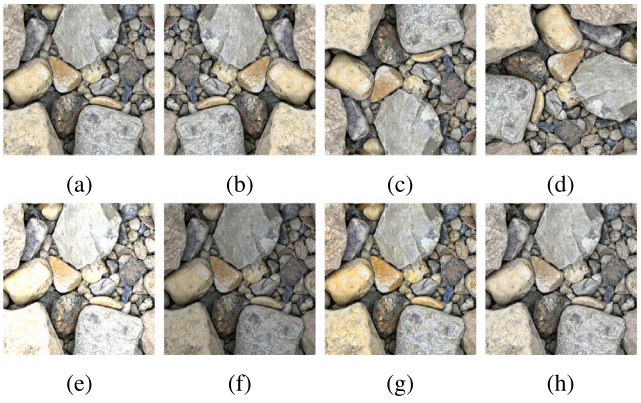


Fig. 7. Example image transformations to augment the data. (a) Original image; (b) horizontal flip; (c) vertical flip; (d) 90° rotation; (e) brightness increase; (f) brightness decrease; (g) saturation increase; (h) saturation decrease.

2.2. Data splitting and augmentation

We partitioned the dataset into a training set (90%) and a held-out test set (10%). The split shown in Fig. 6 reveals that portions of the test distribution lie in regions of label space with sparse training coverage, particularly near $(3, 3)$. This shift implies that the model must extrapolate beyond the densest training regions, providing a more realistic assessment of generalization capability.

To improve robustness, augmentation strategies that preserve semantic content are applied during training (Fig. 7). These include horizontal and vertical flips, 90° rotations and brightness/saturation adjustments (both increase and decrease). During training, a deterministic staged duplication pipeline is applied on the training folds only, yielding multiple variants per original image. Validation and test sets are not augmented to maintain rigorous evaluation standards. After augmentation, the training set comprised 25,728 images.

2.3. Model architecture and training

We predict $\ln(b)$ and c using an EfficientNetV2-S (Tan and Le, 2021) backbone. EfficientNetV2-S is a compact variant of the EfficientNetV2 family that blends MBConv and fused-MBConv blocks with progressive

Table 4
Model architecture: EfficientNetV2-S (Tan and Le, 2021) without classification layer and stacked, trainable neural network.

EfficientNetV2-S				
Stage	Operator	Stride	Channels	Layers
0	Conv3 × 3	2	24	1
1	Fused-MBConv1, k3 × 3	1	24	2
2	Fused-MBConv4, k3 × 3	2	48	4
3	Fused-MBConv4, k3 × 3	2	64	4
4	MBConv4, k3 × 3, SE0.25	2	128	6
5	MBConv6, k3 × 3, SE0.25	1	160	9
6	MBConv6, k3 × 3, SE0.25	2	256	15
Stacked trainable neural network				
Stage	Layer	Neurons	Activation	Dropout
7	Dense	2048	Relu	0.1
8	Dense	512	Relu	0.1
9	Dense	256	Relu	0.1
10	Dense	128	Relu	0.1
11	Dense	2	Linear	–

training to deliver strong accuracy at low computational cost. The backbone was initialized with ImageNet (Deng et al., 2009) pre-training (transfer learning), its final classification layer removed and a feed-forward regression head with architecture 2048-512-256-128-2 was stacked for regression (Table 4).

The full model contains 24,168,418 parameters, of which 3,837,058 are trainable. We optimize them by minimizing mean squared error (MSE) with the Adam optimizer. Early stopping and model checkpointing are employed to mitigate over-fitting. Training uses a mini-batch size of 32.

To assess robustness, we perform 5-fold cross-validation with group-aware splits to prevent leakage of replicate images from the same soil sample across folds. After training, we report the coefficient of determination (R^2) and mean absolute percentage error (MAPE) on both the training and test sets.

2.4. Uncertainty quantification via Monte Carlo dropout

To provide a measure of confidence in the model's predictions, we implemented Monte Carlo (MC) Dropout (Gal and Ghahramani, 2016) as a form of approximate Bayesian inference. Standard neural networks provide point estimates without quantifying the underlying epistemic uncertainty. By keeping dropout layers active during inference and performing 50 stochastic forward passes for each image, we obtain a predictive distribution for the Weibull parameters.

The predictive mean $\bar{y}$ is taken as the final prediction, while the predictive standard deviation σ serves as a proxy for uncertainty. To ensure the reliability of the uncertainty estimates, we performed a post-hoc calibration to scale the raw standard deviations such that the empirical coverage matches the nominal 95% confidence level (see Table 5).

2.5. Comparison with segmentation

We compare GRAI3 with segmenteverygrain (Sylvester et al., 2025), a Python package that detects grain-like objects using a patch-based U-Net (Ronneberger et al., 2015) first pass to generate prompts for the Segment Anything Model (SAM) 2.1 (Ravi et al., 2024); many of its training images come from the sedinet project (Buscombe, 2020). For evaluation, both methods are run on the same hold-out set.

Table 5
Confidence intervals (95% CI) by soil type.

Fig.	Count	b			c		
		True	CI low	CI high	True	CI low	CI high
10(a)	10	0.504	0.345	0.378	2.045	2.063	2.211
10(b)	10	21.094	20.990	26.082	3.162	2.649	2.807
10(c)	10	20.989	20.405	24.471	3.121	2.580	2.776
10(d)	10	29.964	18.434	23.240	3.151	2.363	2.733
10(e)	5	3.328	2.541	7.086	0.520	0.704	0.790
10(f)	3	0.697	0.326	0.501	1.895	1.498	2.201
10(g)	3	0.334	0.345	0.446	1.678	1.708	1.901
10(h)	5	0.051	0.053	0.082	0.844	0.787	1.189
10(i)	9	17.382	10.667	16.597	0.548	0.711	0.794
10(j)	4	36.018	9.906	18.428	0.999	0.751	0.894

3. Results

Training on an NVIDIA GeForce RTX 3060 GPU required approximately 3.5 h. The best checkpoint was evaluated on the full training set and on an external held-out test set. Fig. 8 shows predicted versus observed values for both Weibull parameters. Dashed bands indicate a ±30% tolerance.

For $\ln(b)$, training yielded $R^2 = 0.931$ and MAPE = 18.9%; test performance achieved $R^2 = 0.963$ and MAPE = 23.2%. For c, training yielded $R^2 = 0.918$ and MAPE = 12.58%; test performance achieved $R^2 = 0.880$ and MAPE = 18.6%.

Overall, predictions cluster around the bisector line, indicating good agreement between predicted and observed values. Residuals are larger in regions with sparse training coverage, as expected from the distribution shown in Fig. 6. The slightly better test performance for $\ln(b)$ compared to training suggests good generalization without over-fitting, while the performance on c shows expected degradation on the more challenging test cases. Fig. 9 displays the calibrated uncertainty analysis for both $\ln(b)$ and c on the test set. The left panels show a positive correlation between calibrated uncertainty and absolute error, confirming that the model is "error-aware". The right panels visualize the 95% prediction interval "tunnel" sorted by ground truth values, demonstrating that the totality of test samples fall within the predicted bounds. Fig. 10 shows the resulting PSDs for every soil of the test set. Although the predictions are not always perfect, the estimated percentages of the various soil fractions are reasonably accurate, demonstrating that the model effectively captures the key characteristics of the soil types in the test set.

3.1. Regression versus segmentation

We compared GRAI3 with segmenteverygrain using two representative soils. First, for a very coarse sample (Fig. 10(d), $d_{50} \approx 25$ mm) the segmentation produced by segmenteverygrain is shown in Fig. 11. The segmentation is excellent: extracting maximum grain diameters from that segmentation yields PSD curves (Fig. 12) that agree closely with the ground truth. In this coarse case segmenteverygrain's results are highly consistent across images and, in some instances, even match the ground truth more closely than GRAI3's predictions.

The situation changes markedly for a much finer sample (Fig. 10(e), $d_{50} \approx 2$ mm). As illustrated in Fig. 13, many small grains are missed. We attempted manual tuning (for example, reducing patch size) to improve detection on this image, but these adjustments did not remedy the fundamental failures: the method still missed numerous grains.

3.2. Web application

We developed a Streamlit-based web application (Snowflake Inc., 2026) to deliver the PSD estimation in a user-friendly interface (Fig. 14). The application is hosted as an open-source Space on the Hugging Face platform (Hugging Face Inc., 2026) at the URL https://

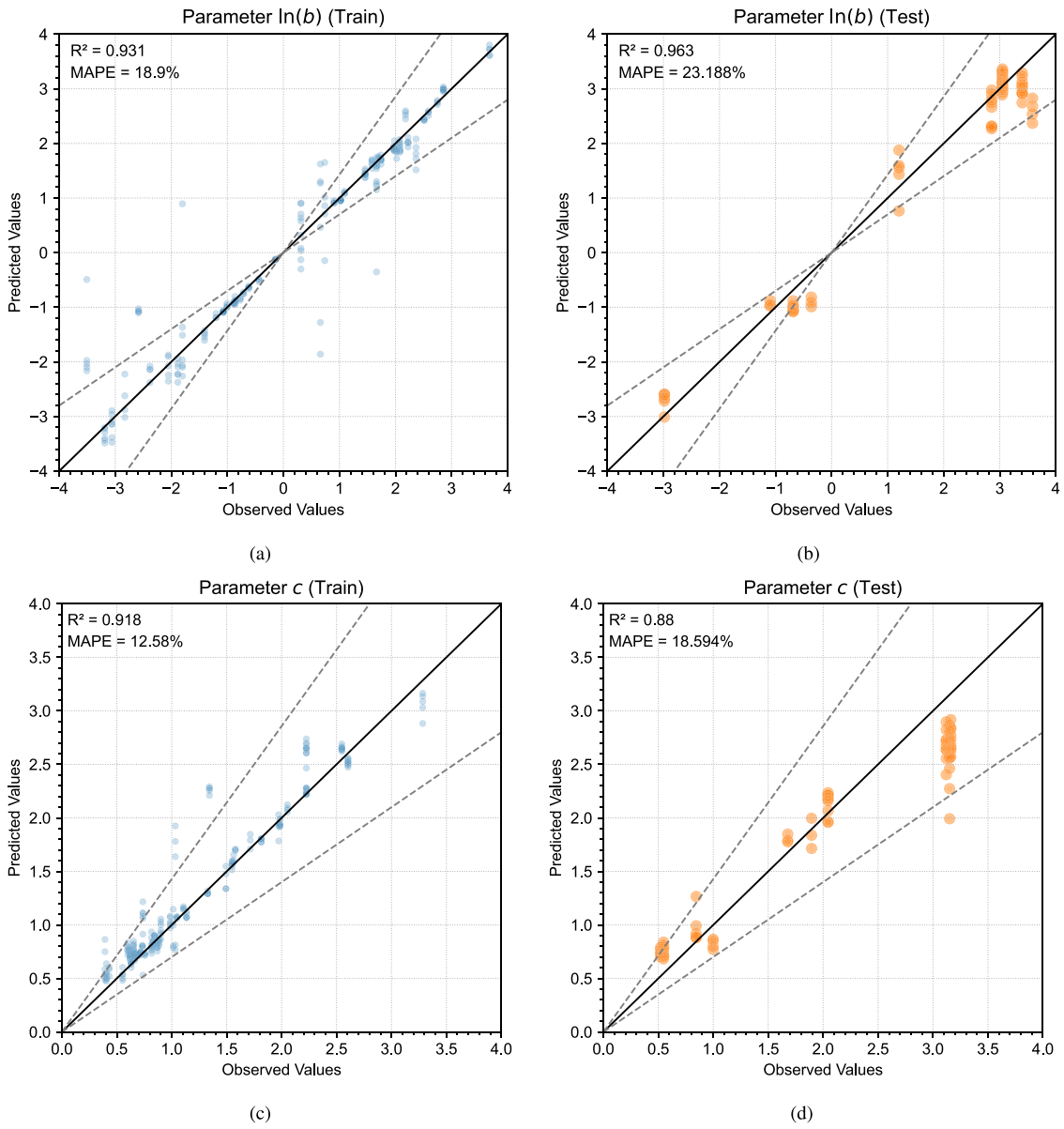


Fig. 8. Prediction performance of Weibull parameters $\ln(b)$ and c for the training and test sets. Dashed lines indicate ±30% tolerance bands around the ideal prediction (solid line).

huggingface.co/spaces/soranz84/grai3, facilitating widespread access for geotechnical practitioners. The pipeline integrates the lightweight compliance classifier to verify image quality before PSD inference.

Upon image upload, the application reads the camera EXIF metadata and attempts to retrieve the pixel density (PPM) from a spreadsheet based on Table 2. If the device is unknown, PPM is estimated based on AOV = 70° and presented to the user for confirmation. All images are normalized to the target pixel density and centre-cropped to the standard input size.

Before PSD inference, a binary CNN classifier gates uploads. The positive class comprises compliant photo-box images of soil; the negative class comprises randomly scraped internet photos (non-soil or irrelevant content). Images with a low soil-compliance score trigger a warning and are blocked unless the user explicitly overrides the decision.

The regression model then predicts $\ln(b)$ and c and the complete PSD curve is computed using Eq. (5) and visualized with standard geotechnical class boundaries (clay, silt, sand, gravel). Derived parameters including D_{10}, D_{50}, D_{60}, coefficient of uniformity (C_U) and coefficient of curvature (C_C) are calculated according to Eq. (6) and

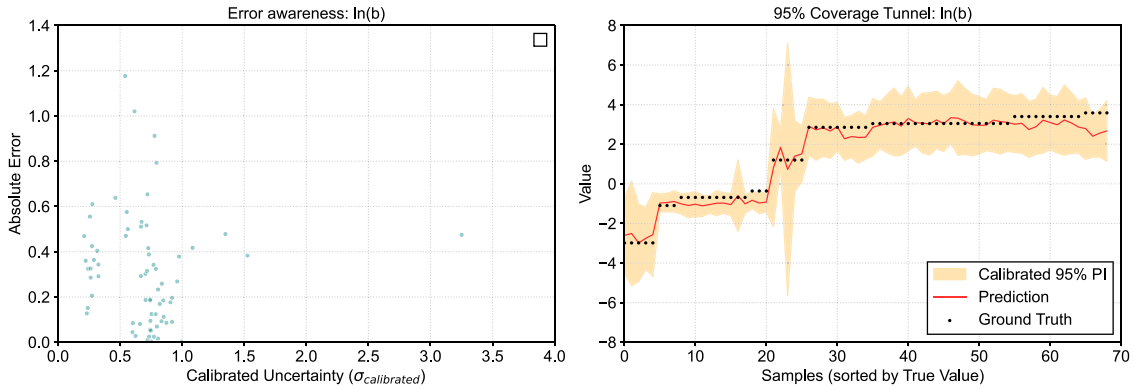
$$C_C = \frac{d_{30}^2}{d_{10} \cdot d_{60}} = \left[\frac{\ln(0.7)^2}{\ln(0.9) \cdot \ln(0.4)}\right]^{1/c} = 1.148^{1/c} \tag{10}$$

and displayed. A one-page PDF report is automatically generated and offered for download, embedding the original photograph, predicted PSD curve, Weibull parameters and derived geotechnical indices. Confidence intervals for b and c can be calculated with Monte Carlo dropout. The workflow of the web application is depicted in Fig. 14.

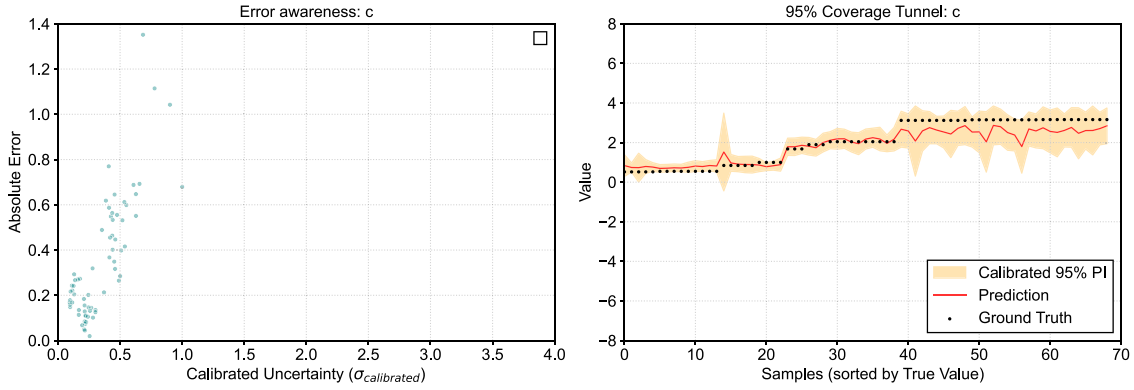
3.2.1. API implementation

To enable programmatic access to the GRAI3 prediction pipeline, we also deployed a dedicated API Space on Hugging Face at https://huggingface.co/spaces/soranz84/grai3API. This Docker-based (Merkel, 2014) API service provides RESTful endpoints for real-time soil particle size distribution prediction, enabling integration into third-party workflows, batch processing pipelines and custom software applications.

The GRAI3API Space is built using a Docker container hosting a Python-based inference server. The API follows the Gradio programming interface paradigm (Abid et al., 2019), which automatically



(a)



(b)

Fig. 9. Uncertainty quantification via MC Dropout for the Weibull parameters: (left) absolute error vs. calibrated uncertainty and (right) 95% prediction interval coverage tunnel for (a) $\ln(b)$ and (b) c.

generates interactive documentation and supports both synchronous and asynchronous inference requests. Upon instantiation, the Space loads the EfficientNetV2-S regression model, the compliance classifier and the calibration lookup table into memory, enabling low-latency predictions without cold start delays.

The API exposes a single primary endpoint that accepts image uploads via HTTP POST requests. Input images are processed through the same preprocessing pipeline as the web application. The regression model then predicts the Weibull parameters $\ln(b)$ and c, from which the full PSD curve and derived geotechnical indices are computed.

Clients interact with the API by sending multipart/form-data requests containing the image file. The API returns a JSON response with the following structure:

- Weibull parameters b and c,
- characteristic diameters and curve shape coefficients,
- soil fractions and
- image dimensions.

The API can be invoked using standard HTTP libraries in any programming language. Example usage in Python:

```
import requests

url = "https://soranz84-grai3api.hf.space/analyze"

with open("soil_image.jpg", "rb") as f:
    response = requests.post(
        url,
        files={"file": ("image.jpg", f, "image/jpeg")}
    )
print("Status:", response.status_code)
print("Response:", response.text)
```

Fig. 15 shows the API implementation in the commercial software GGU-CONNECT (GGU software, 2026). The software links a soil image to a specific segment of a core sample (KDB) and utilizes the GRAI3API to analyse the image, returning the corresponding soil fractions.

4. Discussion and conclusions

By expanding training coverage to poorly graded soils, enforcing group-aware cross-validation and upgrading the backbone to EfficientNetV2-S, GRAI3 achieves stable cross-device performance under harsher evaluation conditions than previous versions. We acknowledge that GRAI relies on global image cues (colour, texture, overall appearance) that are influenced by regional mineralogy and provenance, so its performance may not transfer reliably to geologically distinct soils. The device-agnostic architecture with automatic camera detection and on-the-fly PPM estimation enables operation across virtually any smartphone, significantly broadening the potential user base beyond pre-calibrated devices. We note that while a device-agnostic AOV fallback yields low PPM errors for held-out phones (maximum APE $\approx$ 5%), residual device-dependent normalization errors can still occur for truly unseen hardware, so limited device diversity in the calibration set is a clear limitation and users should consider per-device calibration.

This work assumes the soil surface is approximately flat when applying physical-scale normalization; in practice, coarse gravels and cobbles can create substantial surface relief so the camera-to-surface distance varies across the image, producing spatially varying PPM

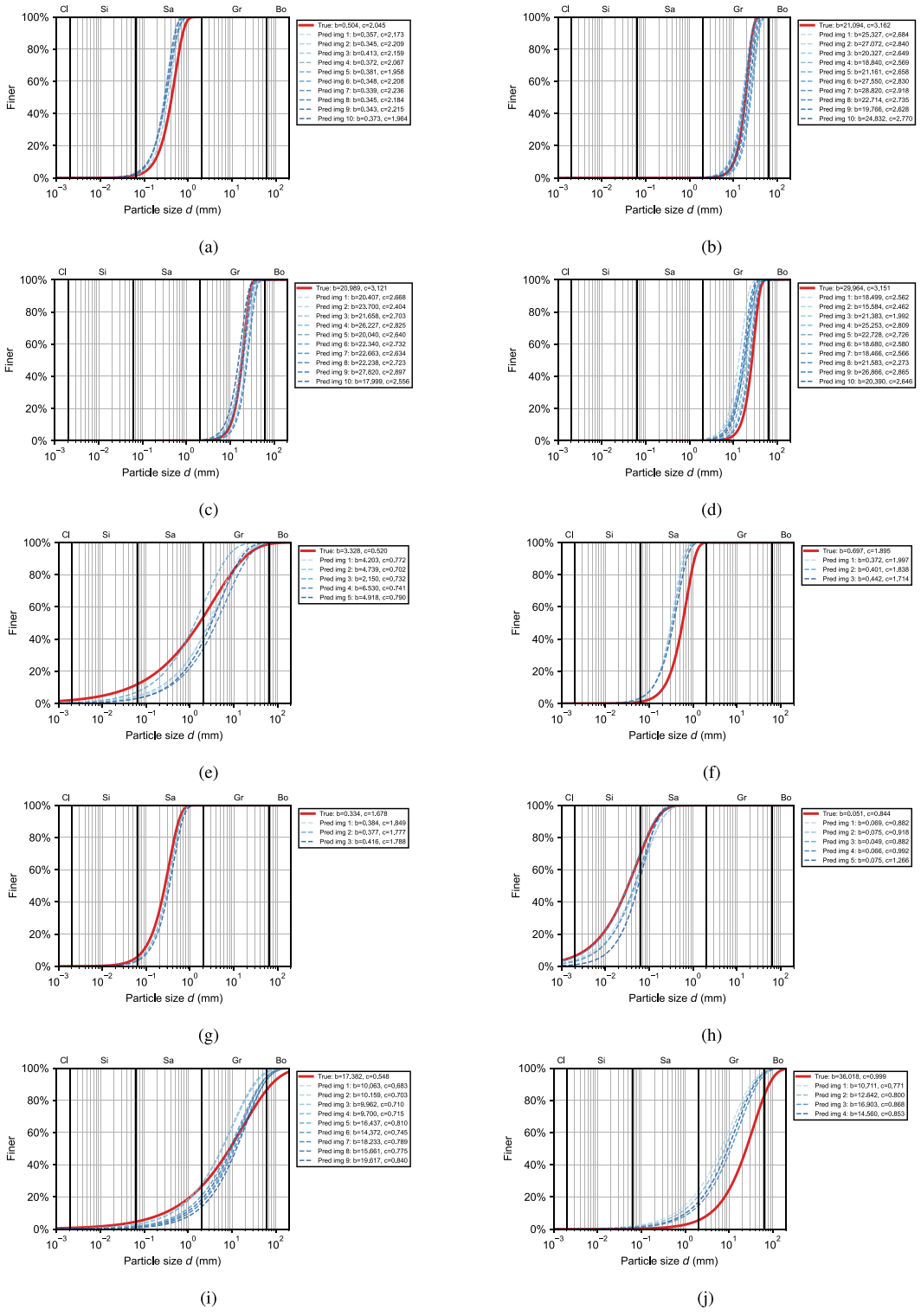


Fig. 10. Results on each sample of the test set. Multiple pictures of each soil were acquired (blue lines) and their predictions were compared to the ground truth (red line).

error. This effect may degrade size estimates for the coarsest fractions and so is a limitation of our method. We note, however, that coarse-grained soils were included in training and the model can still produce reasonable predictions for samples with large gravel/cobble content. The lightweight compliance classifier effectively gates non-soil or non-compliant images, preventing spurious predictions and improving user confidence in the results. We did not perform an extensive quantitative evaluation of classifier accuracy as this work focuses on the PSD regression model; nevertheless, users should be aware that residual non-compliant inputs can degrade downstream predictions.



Fig. 11. Segmented grains obtained with the segmenteverygrain method for the soil with $d_{50} \approx 25$ mm.

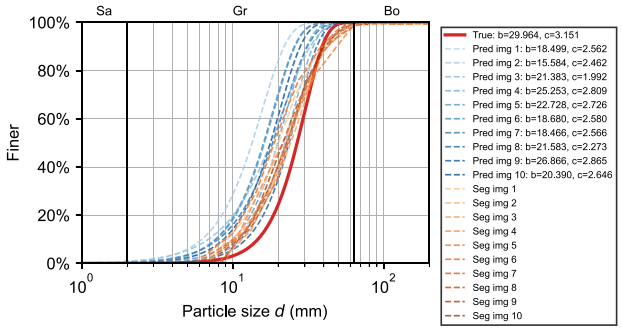


Fig. 12. Comparison of the predictions obtained with GRAI3 and the segmenteverygrain method for the soil with $d_{50} \approx 25$ mm.

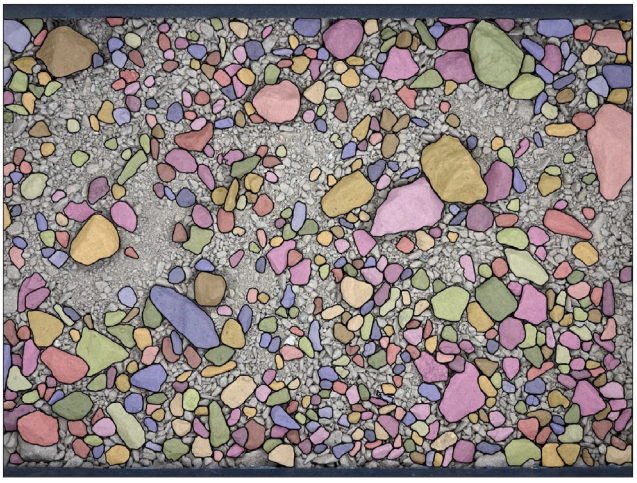


Fig. 13. Segmented grains obtained with the segmenteverygrain method for the soil with $d_{50} \approx 2$ mm.

We compared GRAI3 with a SAM-refined, U-Net-based segmentation pipeline. For coarse materials segmentation can produce very high-fidelity per-grain masks and PSDs that closely match ground truth; however, for finer textures, it misses many grains, degrading PSD recovery even after manual parameter tuning. In contrast, GRAI3's inference remains robust across this size range. The integrated web application provides rapid inference (typically under 2 s per image) and generates comprehensive PDF reports suitable for field documentation and regulatory submissions. These improvements advance image-based PSD estimation from proof-of-concept to a deployable tool suitable for practical geotechnical applications including site investigation, quality control during earthworks and rapid screening of borrow materials.

Performance remains sensitive in sparsely represented regions of label space, particularly for very steep distributions. Representing PSDs by two Weibull parameters enforces a unimodal, smooth shape and therefore cannot capture gap-graded, sharply changing, or strongly multimodal distributions typical of fluvial or lacustrine beds. The current workflow relies on a standardized photo-box and primarily dry, controlled samples; factors common in field settings—moisture can degrade PSD estimates, so the present system should be considered a lab-assisted screening tool rather than a fully robust field solution until further data collection and algorithmic adaptations are completed.

Future research directions include expanding the diversity of tested soil types to improve representativeness across global texture and composition ranges, collecting training data across varying water contents to enable robust field deployment on as-sampled materials, incorporating compositional or spectral features to better capture gap-graded and bimodal PSDs, refining the loss function to penalize misprediction of the fines fraction more heavily (given its disproportionate impact on hydraulic and mechanical behaviour) and studying sedimentation effects and optimal height of soil in the box for expected mean particle size.

GRAI3 represents a significant advancement in smartphone-based soil characterization, addressing key limitations of previous versions through improved model architecture, rigorous validation and device-agnostic design.

1. The system achieves strong predictive performance ($R^2 > 0.88$ on unseen test data) while maintaining practical usability through automatic device calibration and one-click reporting.
2. The web-based implementation lowers barriers to adoption by eliminating the need for specialized software installation or device pre-calibration.
3. Combined with the standardized photo-box protocol, GRAI3 offers a reproducible workflow for rapid PSD estimation that complements traditional laboratory methods, particularly in scenarios requiring quick turnaround, field screening or documentation of spatial variability.
4. While challenges remain for certain soil types and field conditions, the system demonstrates sufficient accuracy and robustness for many practical geotechnical applications.
5. Continued expansion of the training dataset and refinement of the architecture will further enhance generalizability and broaden the range of applicable scenarios.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This work was funded by the European Union under the MSCA Staff Exchanges project 101182689 Geotechnical Resilience through Intelligent Design (GRID). Views and opinions expressed are however those of the author only and do not necessarily reflect those of the European Union or European Research Executive Agency (REA). Neither the European Union nor the granting authority can be held responsible for them.

The author would like to thank Anand D'Souza and Nicolas Huepe-Marinovic of BOKU, Stefan Resch, Lukas Krödel and Dennis Tegen of HPC, Jürgen Matzinger of Geotechnik Tauchmann for providing image data and particle size distributions to train the models; Simon Bußand Daniel Schöler of GGU for integrating GRAI into the GGU-CONNECT software.



Fig. 14. Workflow of the GRAI3 web application.

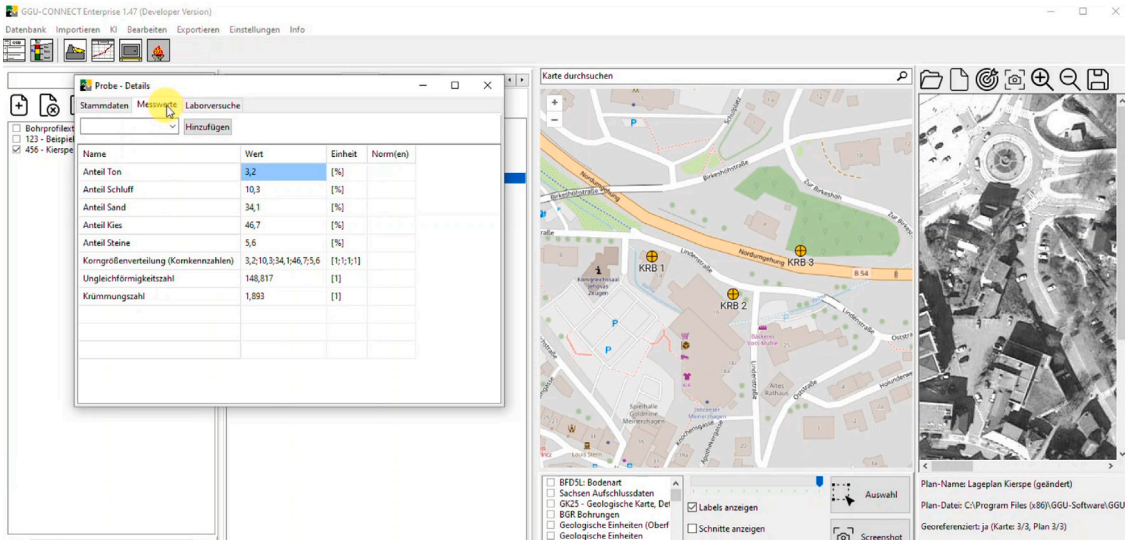


Fig. 15. API implementation in GGU-CONNECT (GGU software, 2026) showing the results obtained from a soil picture associated with a core sample. The results show the various soil fractions (from clay to boulders), the coefficients of uniformity and curvature.

Data availability

Data will be made available on request.

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